

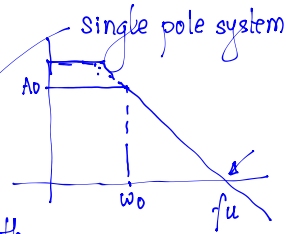
1. Gain

2. Bandwidth:-

$$\omega_u = A_o \omega_o = \text{GBW}$$

Gain

Bandwidth



3. Output voltage swing

4. Linearity

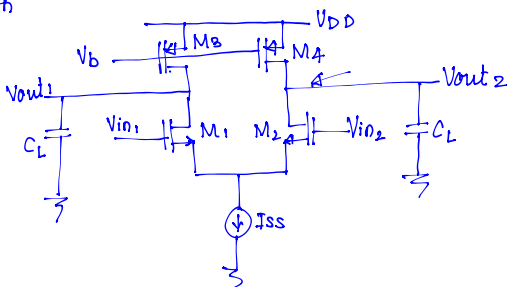
5. Noise

7. Offset :-

One-stage op-amp :-

$$V_{DS} \geq V_{GS} - V_{th}$$

$$V_D \geq V_G - V_{th}$$



$$A_v = g_{m1,2} (r_{o1,2} || r_{o3,4})$$

$$V_{in,cm,min} = V_{SS} + V_{GS1}$$

$$V_{in,cm,max} = V_{DD} - |V_{OV3}| + V_{thn} \quad \checkmark$$

$$V_{out2,max} = V_{DD} - |V_{OV4}|$$

$$V_{out1,max} = V_{DD} - |V_{OV3}|$$

$$V_{out2,min} = V_{SS} + V_{OV2}$$

↑
minimum possible value

$$V_{out1,min} = V_{SS} + V_{OV1}$$

$$V_{out} = V_{out1} - V_{out2}$$

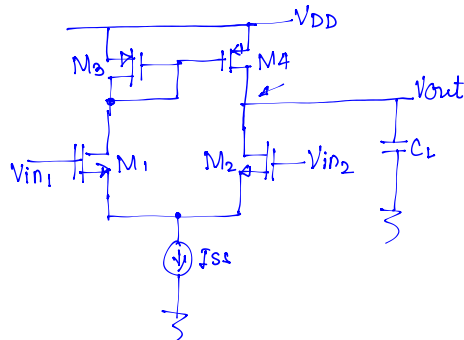
$$V_{out,max} = V_{out1,max} - V_{out2,min}$$

$$V_{out,min} = V_{out1,min} - V_{out2,max}$$

$$\text{Output voltage swing} = V_{out,max} - V_{out,min}$$

$$= 2(V_{DD} - |V_{OV3}| - |V_{OV1}| - V_{SS})$$

$$BW = \omega_{p1} = \frac{1}{(r_{o2} || r_{o4}) C_L}$$



$$A_v = g_{m2} (r_{o2} || r_{o4})$$

$$V_{in,cm,min} = V_{SS} + V_{GS1}$$

$$V_{in,cm,max} = V_{DD} - |V_{GS3}| + V_{th1}$$

CMIR

$$V_{out,max} = V_{DD} - |V_{OV4}|$$

$$V_{out,min} = V_{SS} + V_{OV2}$$

$$\text{Output voltage swing} = V_{out,max} - V_{out,min}$$

$$= V_{DD} - |V_{OV4}| - V_{OV2} - V_{SS}$$

$$BW = \omega_{p1} = \frac{1}{(r_{o2} || r_{o4}) C_L}$$